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Automatic reconstruction of 3D building models from scanned 2D floor plans

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ABSTRACT

The overall energy efficiency of existing buildings has to be significantly improved to comply with emerging regulations and to contribute to overcoming current environmental challenges. Many policies aim at accelerating the renovation rate. The effectiveness of renovation actions could be significantly improved through the systematic use of Information and Communication Technologies (ICT) tools and Building Information Modeling (BIM). But these solutions rely on full-fledged digital models, which, for most buildings, are not available. The present article introduces a research work aiming at the development of methods for the generation of 3D building models from 2D plans. The developed prototype is able to extract information from 2D plans and to generate IFC (Industry Foundation Classes)-compliant 3D models that include the main components of the building: walls, openings, and spaces. The article also presents the results of a quantitative assessment of the platform capabilities and performances, relying on a database of 90 real architectural floor plans. The results are very promising and show that such solutions could be key components of future digital toolkits for renovation design.

1. Introduction

In Europe, a large number of buildings have to undergo major renovation in order to comply with current regulations and environmental challenges. The European building stock includes around 40% of buildings erected before 1960 and the turnover is low (only 1% every year) [1]. Therefore, significant energy efficiency improvement can only be achieved through massive renovation strategies, implemented at National and European level [2]. The BIM and the related software support (e.g. energy simulation tools) help the designers to make better-informed decisions, and result in more optimal, more energy-efficient designs. They could significantly improve the effectiveness of renovation designs and, if systematically used, help to reach the goals set by contemporary environmental issues. This statement is confirmed by recent researches that have demonstrated that advanced ICT support, such as decision-support systems for renovation action selection and assessment, can bring significant benefits in terms of cost and energy efficiency [3].

While BIM ICT tools are undoubtedly beneficial to building design practices and, are now commonplace in new building design processes, they are still under-exploited in renovation projects. Indeed BIM is widely acknowledged as the basis of modern design and brings significant benefits, far beyond visualization and CAD-based design, spanning e.g. seamless design data flow and management [4,5]. However, such advanced design approaches require full-fledged semantized 3D digital models in order to realize their full potential. In most cases, such models are not available for existing buildings. Therefore, one critical short-term research challenge, in the scope of renovation, is to devise effective and reliable methods and tools to reconstruct 3D digital models of existing buildings. Some approaches already exist (e.g. laser scanning, photogrammetry) but the related costs (devices, workload) hinder their wider take-up. The difficulty of creating 3D models at reasonable costs is actually the main factor that still prevents the intensive use of BIM in renovation. Our motivation for the research presented in this paper stems from the recognition of this shortcoming, and aims at contributing to deliver cost-effective and reliable solutions for the reconstruction of 3D BIM of existing buildings.

This work has been initiated by a state-of-the-art review we recently lead [6], which gave an overview on techniques used to generate 3D building models. The main conclusion of this review was that there is currently no general answer about the best approach to generate 3D models of existing buildings: the selection of a method cannot be considered independently from the end user's objectives and the project constraints. The review also highlighted the breadth of the area, and the numerous techniques that aim at the creation of 3D models of existing buildings [7]: photogrammetry [8], laser scanning [9], tagging, use of preexisting information like sketches and tape measurers. But, despite this significant offer, no approach seems to allow for the generation of exploitable 3D building models, that would truthfully represent the geometry, topology and semantics of the buildings. For example, laser scanning results in detailed geometric information but falls short

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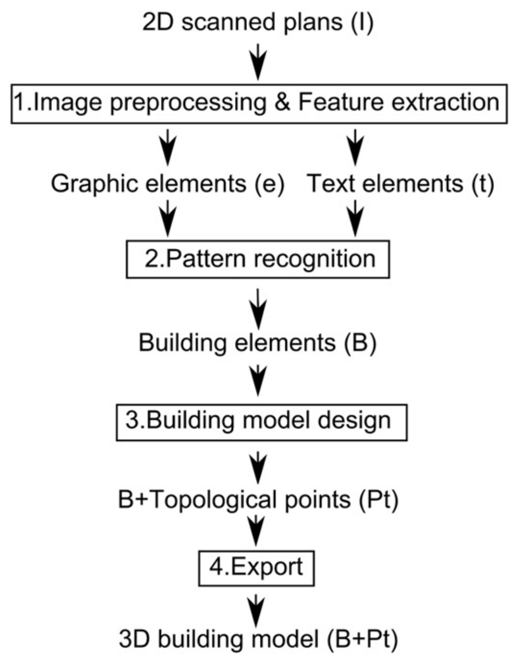

Fig. 1. General process to convert a 2D scanned plan into a 3D building model.

in extracting topology and semantics. The issue is that, to be considered as valid and complete, 3D digital models must include these three components: *geometry*, which defines shape and dimensions, *topology*, which defines the relationships between building components and, *semantics*, which describes additional characteristics, such as room function, usually through dedicated attributes.

Based on this initial assessment, the aim of our work has then been to devise, implement, and test a solution to generate 3D models of existing buildings at reasonable costs and, relying on easily accessible data. We acknowledged the creation of 3D models from 2D scanned plans as the best trade-off between cost and precision. This is an approach that relies on available data and which, does not require any on-site intervention. Moreover, the review highlighted that despite significant advances in document analysis domain, no available solution of this kind featured the whole process, from 2D plans processing to 3D BIM generation. One obvious shortcoming of these methods is that the quality of the generated 3D models highly depends on the quality of the input 2D plans. They can result in incomplete or incorrect models, when the 2D models have not been updated. This has led us to consider the combination of such 2D drawing based-approaches with computer-aided, semantic-aware manual correction. This article presents the design, development and testing of such a solution, starting with the description of the process developed to convert a 2D scanned plan into a 3D building model in Section 2. The subsequent section (Section 3) outlines the results of a validation based on a corpus of 90 architectural floor plans. Section 4 discusses the results obtained and highlights envisioned improvements, before concluding.

2. Overview of the 3D model creation process

In this section, the general process to convert a 2D scanned plan into a 3D building model is presented (Fig. 1). It starts with the generation of an architectural floor plan image from a 2D paper plan thanks to an optical imaging scanner. Then, three successive processing phases are defined as follows. The first (see Section 2.1) deals with preprocessing and extraction of image features, i.e. identification of the geometrical primitives and separation of texts and graphical elements). The second phase (Section 2.1) deals with the recognition of building elements graphical patterns (walls and openings in our case). At last, the building model is generated (Section 2.3).

2.1. Feature extraction

Prior to addressing graphical feature extraction, it is necessary to pre-process the image in order to clean it and remove useless information. This first preprocessing step consists in binarizing the image to remove potential noises caused by the original paper plan quality or by the scanning itself (step 1 of the process). In what follows, the original image will be noted *I*.

The second step of this first phase is focused on feature extraction, to separate geometrical primitives from text elements. This separation will then allow to adapt the subsequent processing to the type of the graphical element considered. To this end, the input image is separated into two images: the text image containing all textual information (annotations, title, room names) and the geometry image containing all geometrical elements (lines, circles). This separation is performed thanks to freely available tools from the image processing and document analysis domains [10] and is independent from text font, size and orientation. Once these two images are generated, they are further processed separately. The text image is analyzed thanks to OCR (Optical Character Recognition) techniques to identify text items and associated bounding boxes. This analysis relies on the tools developed by the Qgar project (Loria laborataries, France) that allow extracting sets of characters in the image and their bounding boxes. A bounding box is a rectangle defined by its diagonal that surrounds a given set of characters. Prior to defining a text element, it is necessary to give the definition of a point *P*.

$$P = (x, y) \tag{1}$$

A point is defined by two coordinates which correspond to a pixel in the image *I*. A text element (t) is therefore defined as:

$$t = \{P_1, P_2, textString\} \tag{2}$$

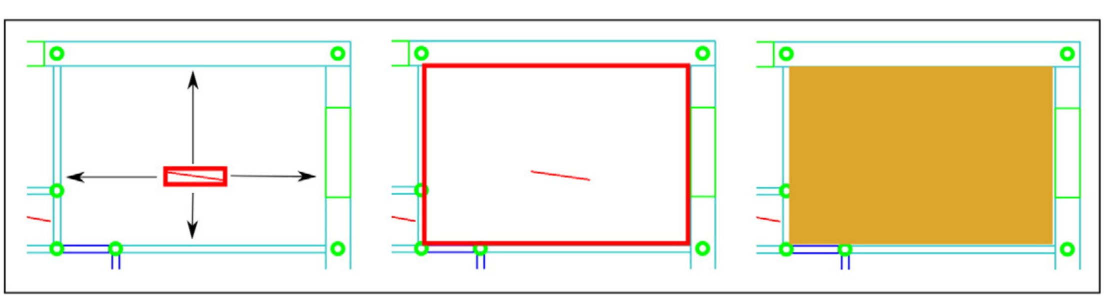

Fig. 2. Space reconstruction: case of a space representing by a rectangle. In the left image, the red segment represent a text element surrounding by its associated bounding box. The bounding box is extended along 4 directions. Green points represent the set of topological points. The middle image represents the region defined by the extension of the bounding box and the yellow polygon is the final identified space.

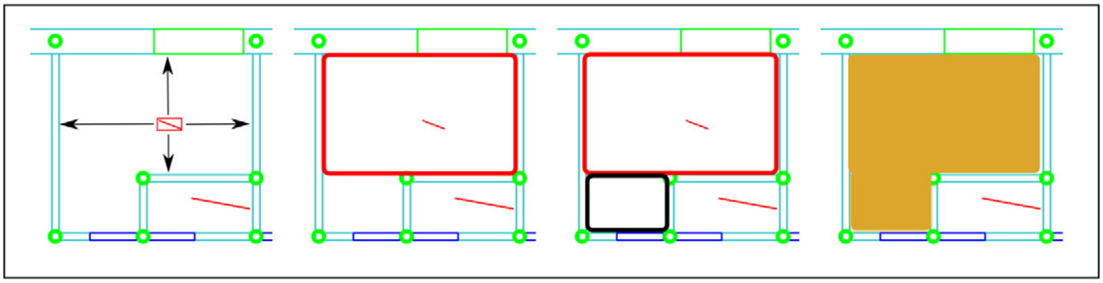

Fig. 3. Space reconstruction: case of a complex shape.In the left image, the red segment represent a text element surrounding by its associated bounding box. The bounding box is extended along 4 directions. The red rectangle is the next image is the region defined by the extension of the bounding box. In this case, this region is insufficient to determine the space. The black rectangle represent the research and the addition of a complementary region to the first one. The yellow polygon in the right image is the final identified space.

where P_1 and P_2 are the two points at the tips of the diagonal of the bounding box. These two points are at the top-left and bottom-right corners. TextString is the word recognized in the image *I*.

As for the geometry image, the focus is first on the identification of geometry primitives (e) which is a set of points that can be for example a segment or an arc, is defined as follows:

$$e = \{P_i\}_{i=2..n} \tag{3}$$

Two different geometrical shapes are searched for in the graphical image: segments and arcs. The assumption made here is that these two primitives are enough to represent all the building components targeted (walls, openings). Also, while we recognize the variety of graphic conventions applying to architectural floor plans design, we take the assumption that in most cases a wall will be represented by a black or white rectangle. As regards the geometry primitive identification, segment and arc detection is performed, thanks to Hough methods [11].

The segment extraction method is derived from the standard Hough method to detect a line. When a line is detected, the longest segment composed of continuous pixels or with a gap not exceeding a given threshold is searched [12]. The circle extraction is also based on Hough method but using a circle parametrization which is given by the center's coordinates and the radius. Any given edge point could be a point of a right circular cone in the parameter space. Then if the cones corresponding to many edge points intersected at a single point, the three parameters for the circle are found [13].

2.2. Pattern recognition

Using the set of previous extracted text elements (t) and geometry elements (e), the next step is to define building elements (B). A building element is defined by:

$$B = \left\{ \{e_i\}_{i=0..n}, \{t_j\}_{j=0..p}, \{B_k\}_{k=0..q} \right\} \tag{4}$$

It is important to note that a building element can be composed of other building elements. The two following subsections present two classes of building elements: wall (B_w) and opening (B_o).

2.2.1. Wall segments recognition

Wall recognition is a widely addressed topic. This step is critical to the recognition stage, particularly to determine the building footprint. One of the main difficulties is to differentiate between the lines that represent other building components (e.g. lines which represent the roof) and the lines that represent walls. Some approaches made the assumption that thin and thick lines shall be separated before performing wall search and that thicker lines represent the walls [14]. However, this assumption is too restrictive in our case and cannot be applied to various plan representations. A wall — or more precisely, a wall segment — (B_w) is therefore defined as follows:

$$B_w = \begin{cases} e_1, e_2 \\ e_{1_{shape}} = e_{2_{shape}} = segment \\ e_1 // e_2 \\ S_{min} < d(e_1, e_2) < S_{max} \end{cases} \tag{5}$$

This equation basically tells that a wall segment is defined by a couple of parallel lines, which are separated by a distance $d(e_1, e_2)$ that lies in between two predefined values. These standard values (S_{min}, S_{max}) have been set according to the drawing conventions of the plans used as part of our study. These conventions seem not to vary much and it is likely that the values as configured will be suitable for most cases. They are however configurable and shall be tuned if necessary. In our approach, the identification of wall segments is also the opportunity to perform a first round of correction actions. All wall segments are checked against additional validation rules to eventually correct or harmonize thicknesses. Thickness values are automatically determined by analyzing all B_w thicknesses. An automatic process allows to define the number of thickness classes and to find the most appropriate value. Then, each B_w thickness is harmonized using the original image to validate the thickness and to eventually correct the wall position. The set of B_w is split into two groups: one for outdoor walls (B_{wO}) and one for indoor walls (B_{wI}).

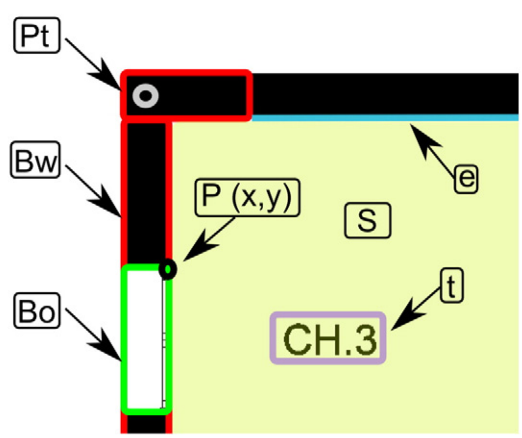

Fig. 4. Example of extracted features from an original image and recognized building elements. A segment is represented in blue, a point in black and a text element in purple. Two walls are represented in red with a topological point in gray and an opening in green. The yellow area correspond to a space.

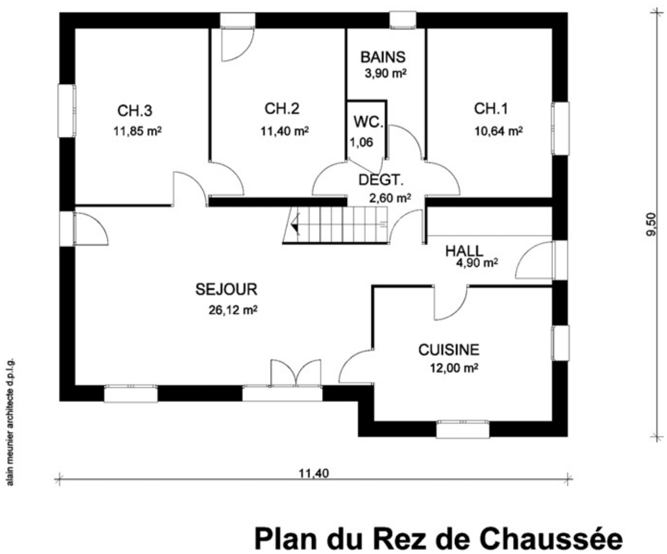

Fig. 5. Architectural floor plan example: original image.

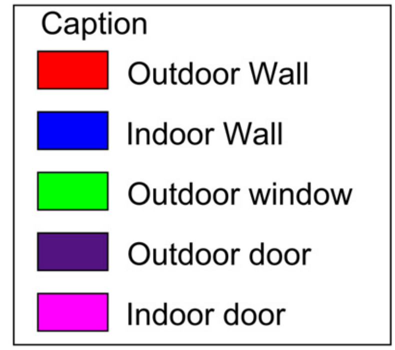

Fig. 7. Caption used to represent recognized building elements in all illustrations.

2.2.2. Openings recognition

The representation of *opening* depends on the drawing conventions that apply, but is represented in most cases by a rectangle included in a wall. In our study, the following assumptions were made: (i) each opening is linked to only one wall, (ii) an opening can be a door or a window. Relying on these assumptions, an opening building element (B $_o$) is defined as:

$$B_o = \begin{cases} e_1, e_2, B_w, (e_3) \\ e_{1_{shape}} = e_{2_{shape}} = segment \\ e_{3_{shape}} = arc \\ e_1 \subset B_w \\ e_2 \subset B_w \\ \left| d(e_1, e_2) - d\left(B_{w_{e1}}, B_{w_{e2}}\right) \right| < \epsilon \end{cases} \tag{6}$$

where e_1 and e_2 are two segments in intersection with the wall B_w and the distance between e_1 and e_2 is similar to the thickness of B_w. So the difference ϵ between the two thicknesses should be low. An opening can be eventually represented with an arc defined by e_3.

Not all of the plans are designed following the same drawing conventions but an opening is usually positioned inside its associated wall. So, we can identify outdoor openings by looking for discontinuities in the wall representation. A comparison between the original image and previous identified B $_w$ allows to recognize openings. To distinguish a door from a window, an arc can be drawn to represent a door, explaining why parameter e3 is optional.

2.3. Building 3D model creation

2.3.1. Building outer shape creation

The first step of the building 3D model creation is to identify in the set of B $_w$ elements the ones that correspond to outdoor walls. Then, the aim is to generate from these outdoor walls a complete and valid outer shape. This step is essential because other building elements will be positioned using this footprint as a reference. The outer shape of the building is considered valid when it is composed of a set of outdoor walls forming a closed shape geometrically and topologically. The rationale of the outer shape reconstruction algorithm is to look for intersections between outdoor walls and to create topological points at each of them. A topological point (P_t) is defined as:

$$P_t = (B_{w1}, B_{w2}, (x, y)) \tag{7}$$

where B_{w1} and B_{w2} are the wall that intersect, and (x,y) are the coordinates of the topological point.

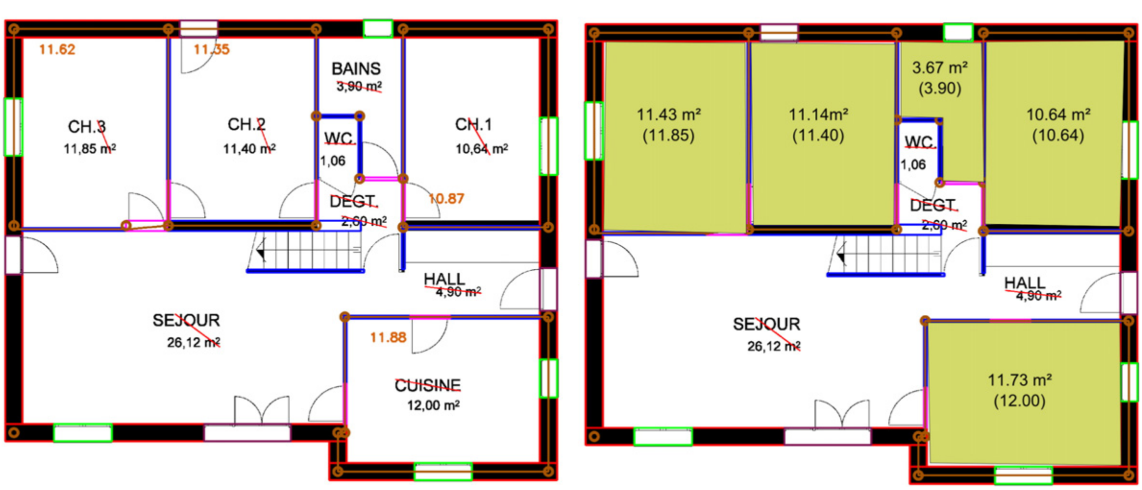

Fig. 6. Result of the automatic recognition: the left image represent the building elements recognizing using the caption of the Fig. 7. The right image represents identified spaces. The shape area in brackets refers to the indicated shape area in the floor plan.

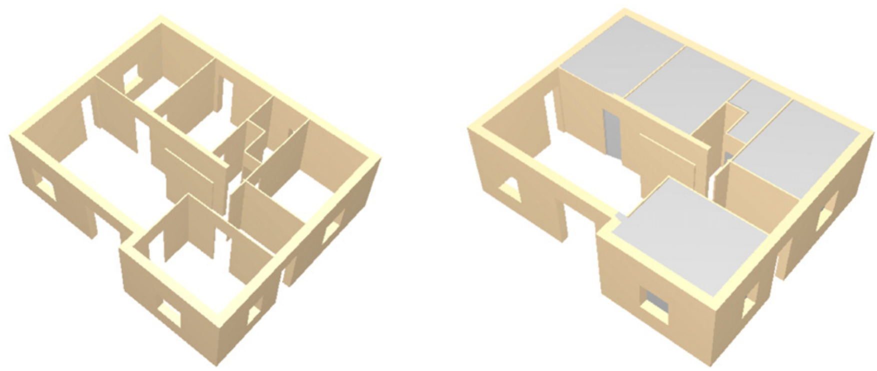

Fig. 8. Visualization of the exported model in ifc format. The right image represents by gray boxes the identified spaces. The software used to visualize this models is DDS-CAD viewer.

To be part of the outer shape, an outdoor wall has to be topologically linked with another outdoor wall at its two tips. The outer shape, or building footprint (F_B) is therefore defined as:

$$F_B = \begin{cases} \{B_{wO_i}\}_{i=3..n}, \left\{B_{o_j}\right\}_{j=1..p}, \{P_{t_k}\}_{k=3..q} \\ \forall B_o \in \left\{B_{o_j}\right\}_{j=1..p}, B_{o_{B_w}} \in \{B_{wO_i}\}_{i=3..n} \\ \forall B_w \in \{B_{wO_i}\}_{i=3..n}, \exists!(P_{t1}, P_{t2}) \in \{P_{t_k}\}_{k=3..q} \end{cases} \quad (8)$$

where each wall B_w in F_B is also part of two topological points (P_{t1},P_{t2}). Each opening B_o has to be part of a B_w also in F_B. To generate a closed footprint of the building, it is necessary to have at least three building walls and three topological points.

2.3.2. Indoor wall and opening creation

Once the outer shape is created, the subsequent step is to create indoor building elements. An *indoor wall* is a wall which is not part of the outer shape of the building but that is located inside it. An indoor wall can be topologically linked with outdoor or indoor walls anywhere. An *indoor opening* is an opening that is part of an indoor wall. Methods and criteria used for indoor building element identification are different from those applied to outdoor element identification. The method is based on wall intersection detection and on subsequent adjustment of wall dimensions. The algorithm also includes techniques to find missing openings, in particular in the cases where gaps between walls could correspond to an opening.

2.3.3. Building space identification

Using topological and geometrical information generated from the previously described building element recognition steps and, relying on textual information, spaces can then be located and created in the building model. A space is a homogeneous zone whose perimeter is defined by a set of walls and openings. It is supposed to be in some way autonomous from other spaces and, possibly to relate to a specific usage. In particular, the division of the whole surface areas into distinct spaces can facilitate the creation of thermal zones for energy simulation purposes. We define a space (S) the following way:

$$S = \begin{cases} \{P_{t_k}\}_{k=3..n}, t_1 \\ t_1 \subset S \\ A_S > A_{min} \end{cases} \quad (9)$$

A space is composed by a text element t_1 (assumed to correspond to the name of the space), a set of walls forming a closed shape and that includes at least one opening. Moreover, the area of the space A_S has to be higher than a predefined standard value A_{min}. The information given by the text element is very useful because a space is generally defined by its name in a plan. It also provides semantic information to determine the usage of the considered space and to infer additional characteristics. In most cases, a given space will be associated to several text bounding boxes. Should it happen, the prototype automatically select the ones that relate to the same space and merge them (e.g. the name of a space "kitchen" and its associated area "12 m^2").

To summarize, spaces are characterized on the one side by the wall topological graph and, on the other side, by the text associated to them. The wall topological graph is composed of the nodes (or topological points P $_t$) generated during the pattern recognition step, each node representing a relationship between two walls.

Our proposed method is similar to existing methods that are based on the search of walls surrounding a text element [15]. However, in our approach, walls are recognized prior to the space reconstruction based on text elements. The latter — that relies on region's growth methods [16] — is in our approach only one of the to components of spaces reconstruction. The rationale is that each bounding box is extended along four directions until intersecting a wall. This first region generated allows to identify a set of walls belonging to the space. Then,

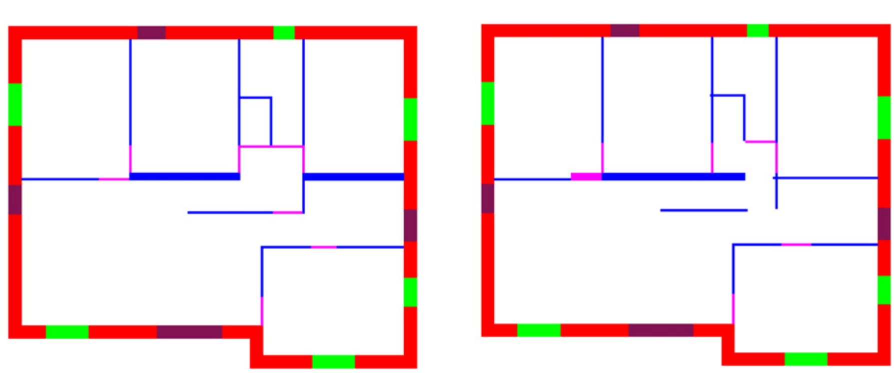

Fig. 9. The image on the left is the ground-truth of the Fig. 5. The image of the right corresponds to the resulting image used in the pixel level evaluation.

using the topological graph, a search is done to determine the closed cycle composed of all selected walls (see Fig. 2). In some cases, it is necessary to search a complementary region in coherence with the first to complete the space (see Fig. 3).

Each space is then validated if at least one opening can be found and by calculating its surface area, which has to be higher than a predefined value. Additional semantic information can then be attached to the space created, thanks to its text elements. Fig. 4 gives an example of extracted features (point, segment, text element) and recognized building elements (wall, opening, topological point and space).

3. Implementation and validation of the 3D models creation solution

In this section, we outline the implementation of a prototype using the methods introduced in the previous sections.

3.1. Presentation of the prototype of the 3D models creation solution

Our software to automatically generate a 3D building model from a 2D scanned plan has been developed in C++. Relying on the results of our recent state-of-the art [6] we have selected the most relevant base components and tools as a starting base. We have used the function developed by the Qgar project[1] led in LORIA (France) to separate the text and the graphic elements during the feature extraction step. The OpenCV library[2] (Open Source Computer Vision Library) is used for image processing: Hough method, reading and writing images. Pattern recognition and 3D building model design modules were developed in-house, based on the algorithms and rules described in the previous sections. Some parameters are not hard-coded and can be set by the user: the minimum and maximum thickness values for the feature extraction, the height of the building and the scale.

To evaluate and validate our prototype, we relied on a set of original floor plan images, from the "Universitat Autonoma de Barcelona", that has already been widely used [17]. It includes 90 floor plans from various complexity, details and configurations.

For illustration purposes, Fig. 5 gives an example of an original floor plan used as input data in our validation and testing activities and Fig. 6 displays the result of the automatic recognition. In this figure, the first image shows results concerning wall and opening recognition with a semantic distinction between door/window and indoor/outdoor, while the second image is focused on space recognition. Five spaces have been recognized and their calculated areas are in line with the values indicated on the plan. The legend is given in the Fig. 7. At the end of the process, a file in IFC format (Industry Foundation Classes) is generated and can be visualized thanks to software tools such as DDS-CAD viewer[3] (Fig. 8). The time necessary to generate the 3D building model for this plan is less than 3 min.

3.2. Quantitative evaluation of the validation results

We have applied two metrics for the evaluation of the results of the validation of our tool. The first relies on a pixel-based method. The second one consists in an evaluation based on the building shape and its properties.

3.2.1. Pixel level evaluation

In [17] a method to evaluate results based on a pixel-evaluation is proposed. The idea is to compare results of the recognition using a ground-truth image pixel by pixel. A pixel can have only two values: 1 if it is a wall pixel or 0. Then, the Jaccard Index (JI) is used to present results, it takes into account pixels that are: true positive, false positive and false negative. The Jaccard Index is defined as follows:

$$JI = \frac{TruePositive}{TruePositive + FalsePositive + FalseNegative} \quad (10)$$

Table 1
Evaluation of the 90 floor plan database. Results are expressed using the Jaccard Index.

Global evaluation	0.63
Walls	0.76
Outdoor walls	0.66
Indoor walls	0.40
Openings	0.53
Outdoor openings	0.52
Indoor openings	0.41

In this evaluation, only two building element categories are evaluated: walls and rooms. This method is not suitable to our work, because we also aim at the recognition of openings and distinguish between indoor and outdoor elements. In order to better evaluate our results, we have therefore decided to rely on an evaluation of the reliability would take into account the following building elements: indoor & outdoor walls, indoor & outdoor doors and outdoor windows. We have manually annotated each plan of the database to highlight each the building elements previously quoted. Then, we have evaluated our results using the Jaccard Index according to the following criteria:

1. Global Evaluation
2. Walls without taking into account the distinction between outdoor and indoor
3. Outdoor walls
4. Indoor walls
5. Openings without taking into account the distinction between outdoor and indoor
6. Outdoor openings
7. Indoor openings

The Fig. 9 represents the ground-truth image of the floor plan of the Fig. 5 and the resulting image after the automatic recognition. Differences between the ground truth image and the result can be identified, such as thicknesses of an indoor wall on the right or missing indoor doors. The evaluation has been led on the 90 floors plans of the database and the JI has been calculated (Table 1). The global evaluation value is 0.63 for the 90 plans, wall recognition is 0.76 and opening recognition is 0.53. These results show that many openings are missing. The Fig. 10 represents the distribution of recognition value for the global, wall and opening recognition. It clearly shows the weak results obtained on the recognition of openings and the related impact on global quality.

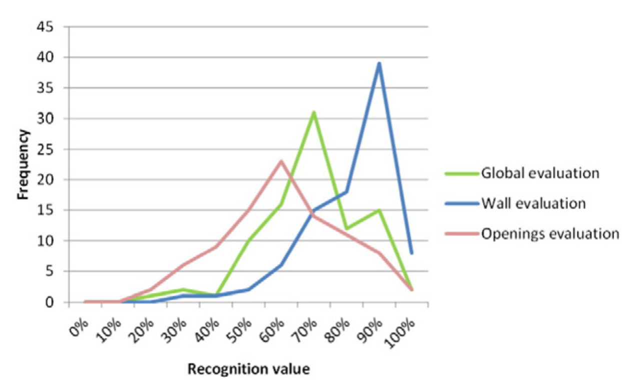

Fig. 10. Frequency of the Jaccard Index value for the global, wall and opening evaluation for the 90 floor plans using the pixel evaluation.

[1] http://www.qgar.org/.
[2] http://opencv.org/.
[3] http://www.dds-cad.net/.

Table 2
Confusion matrix resulting of the evaluation of the 90 plans. In this matrix, the semantic distinction between outdoor and indoor is taking into account.

		Recognition					
		Ground	Ind. wall	Out. wall	Ind. door	Out. wind	Out. door
Ground-truth	Ground	99.67%	0.10%	0.13%	0.01%	0.06%	0.02%
	Ind. wall	17.65%	73.11%	8.18%	0.13%	0.77%	0.16%
	Out. wall	13.03%	13.36%	72.69%	0.00%	0.83%	0.10%
	Ind. door	37.56%	5.16%	1.19%	51.10%	0.08%	4.92%
	Out. wind	31.31%	7.43%	2.87%	0.02%	56.74%	1.64%
	Out. door	23.54%	8.57%	2.12%	3.24%	3.58%	58.95%

To analyze these results more thoroughly, we have also calculated a confusion matrix to measure the quality of the recognition. This matrix gives a clear and synthetic view of the recognition errors. The confusion matrix in Table 2 takes into account the errors made on the semantic distinction between indoor and outdoor elements. It shows that indoor and outdoor walls are recognized in the same proportion (around 73%). However, a significant part of outdoor and indoor walls are not well semantized (13% of outdoor walls and 8% of indoor walls). This confusion impacts in turn the recognition of openings: 3% of outdoor doors are confused with indoor walls and 5% of indoor doors are confused with outdoor doors. Moreover, between 25% to 40% of openings are missing.

It is difficult to compare our results with previous research works because in most cases, the objectives were not the same as ours. Recent results obtained by [17] give a Jaccard Index at 0.97 on wall recognition, which could correspond in our evaluation in Table 1 to the second line related to wall 0.76. However, this comparison is not fully relevant because we did not use the same ground-truth database and our methodology is based on techniques to adjust and harmonize wall representation. For example, we adjust walls when they intersect. This is why some errors are due to pixel precision and little mismatches.

Wall recognition performances have a strong impact on (pixel-based) global recognition rate because we evaluate areas so the error is quadratic. Indeed, an indoor door is usually composed of 700 pixels with a thickness of 7 pixels while an outdoor wall can contains until 50.000 pixels with a thickness of 42 pixels. For example, in Fig. 3 according to the Jaccard index, 56.15% of indoor doors are well recognized, in reality, only 2/9 doors are not recognized and we can say that around 78% of indoor doors are well recognized. This is the reason why we propose an additional evaluation approach based on building characteristics, presented in the following subsection.

3.2.2. Building elements quantitative evaluation

This approach is proposed to evaluate our results at the building-level, and not any more at the pixel level. The aim of our work is actually to provide a 3D building model usable in third-party applications, such as energy simulation. Therefore, the issue is more to ensure that the building is compliant with the applicable geometrical and topological rules, more than guaranteeing pixel-level precision.

The main focus is on walls and openings recognition. Openings recognition performances are evaluated by counting openings that are well recognized, openings that are missing and false openings. For example, in the floor plan of the Fig. 5, 4/4 outdoor doors, 6/6 outdoor windows and 7/9 (78%) indoor doors are well recognized, 2/9 indoor doors (22%) are missing and recognized as ground (no faulty openings in this example). Results for all the plans data base are presented in the Table 3. Without considering the distinction between indoor and outdoor doors, 1080/1738 openings (62%) are well recognized. It is important to note that our solution generates few false openings (101 false openings out of 1839 elements (6%)).

To evaluate walls recognition performances, we propose to count the number of linear meters of walls that are well recognized. For example, in the plan of the Fig. 5 all indoor and outdoor walls are well recognized but one meter of walls is false. The Table 4 corresponds to the evaluation of the 90 floor plans. 6513 m over 7551 (86%) of walls are well identified and similarly to the opening recognition, few false linear meters of walls are identified (274 m out of 7825 recognized meters (3.5%)). Moreover, many outdoor walls are recognized as indoor walls (14%).

4. Discussion

4.1. Performance

Our solution allows to automatically generate in a very short time a 3D building model from a 2D scanned plan (less than 3 min per plan without code optimization). Results obtained are very promising even if some improvements can be envisaged: in the building elements-level evaluation, 63% of openings are well recognized and 86% of linear meters of walls are well identified.

The first step of the process is to determine the outer shape of the building but in some cases, it is not validated (open outer shape, missing wall) and some outdoor walls are converted into indoor walls because they are not topologically linked with outdoor walls that composed the outer shape. This is why, around 14% of recognized indoor walls are in reality outdoor walls. This confusion partly explains why so many outdoor windows are missing (34%): they cannot be identified if the related wall is not an outdoor wall. Moreover, it is important to note that our process generates few false building elements: only 6% of recognized openings and 3.5% of linear meters. Indeed, to correct later the 3D building element it is easier to create a new element than to remove a false.

Our solution allows to tackle the whole process of the recognition and is not just focused on a specific step. Building elements are recognized from the 2D scanned plan and linked together to create a coherent building model. It is then exported in a BIM compliant format (IFC) and can be visualized in BIM software.

Table 3
Results evaluation concerning the number of openings. Results are in number of openings.

	Missing	Outdoor doors	Outdoor window	Indoor doors
False	x	17	60	24
Outdoor doors	84 (29.79%)	170 (60.29%)	14 (4.96)%	14 (4.96%)
Outdoor windows	303 (37.64%)	13 (1.61%)	489 (60.75%)	0 (0%)
Indoor doors	271 (41.62%)	10 (1.54%)	0 (0%)	370 (56.84%)

Recognized openings: 62% — Missing: 38%.

Table 4
Results evaluation concerning the number of linear meters of walls. Results are in meters.

	Missing wall	Outdoor walls	Indoor walls
False	x	167	107
Outdoor walls	707 (14.78%)	3429 (71.68%)	648 (13.54)%
Indoor walls	330 (11.93%)	119 (4.32%)	2316 (83.75%)

Recognized walls: 86% - Missing: 14%.

As already mentioned in the previously section, it is difficult to compare our results with the state-of-the-art because we have used our own ground-truth image and moreover, we do not focus on the same elements in the original image. Our aim is to provide a 3D building model consistent with respect to the three components of 3D models (geometry, topology and semantics), while the other works are more focused on a precise recognition at the pixel level. They do not reconstruct elements, but focus on pixel labeling.

The Table 5 is a qualitative comparison of our solution to three methods of the state-of-the art. The main difference is that we design all the building model by integrating walls, doors windows and spaces. Our underlying internal model is compliant with standard BIM formats and allows exporting the model to third party software. However, oppositely, our approach allows only for the recognition of basic geometries while the other works deal with more complex geometry. A last point relates to the evaluation of our prototype, which could be extended to larger 2D plans samples.

4.2. Limitations

The first limitation of our method concerns the third dimension such as height of the building and openings. In the present version of the software, these values are set manually by the user and the height of all components (openings and walls) is assumed to be equal. Of course, this assumption is not valid on most real cases. Therefore, the tool has to be extended in order to be able to infer these values from other sources.

A second issue relates to tackling the diversity of drawing conventions in architectural plans, which remains a research challenge. The present version of our software relies on assumptions: e.g. only walls composed of straight lines and textured by homogeneous areas of black pixels are identified and doors have to be adorned with arcs. The performances of our tool would therefore be impacted each time new drawing conventions are considered. However, one strength of our process is that the identification process is based on the detection of a change of texture between walls and openings. To our knowledge, this detection pattern is effective with most drawing conventions.

A third limitation is the management of complex architectures - for instance, the tool does not manage curved walls. Some extensions could be implemented to deal with more complex buildings, but this would translate into significant developments. We are not fully convinced, given that most buildings have simple shapes, that such an extension would be worth the additional effort.

At last, space recognition also relies on simplifying assumptions: a space is supposed to be composed of a text element and by polygonal shapes like rectangles. This results in a significant number of unrecognized spaces in the case of complex geometries. For example, in Fig. 6 the “HALL” is not identified because the border of the space is not well defined by walls. Therefore, improving our method of region extension to more reliably identify spaces is part of our perspectives.

Table 5
Qualitative comparison of our solution to methods of the state-of-the-art. Wa = Wall, Do = Door, Wi = Window, S = Space.

Steps	Macé et al. [16]	Ahmed et al. [15]	Heras et al. [17]	Proposed
1. Feature extraction	Line/arc			
2. Pattern recognition	Wa/Do			Wa/Do/Wi
3. Building model design	S	S/(Wa/Do)	S	Wa/Do/Wi/S
4. 3D Export	x	≈	x	✓
5. Evaluation (number of plans)	80	80	122	90

4.3. Envisioned improvements

Our solution is designed to be extendable with external knowledge and data. In this respect, several improvements are envisioned, especially to enhance reliability of the solution. The first issue concerns the third dimension: heights of the building, openings, etc. To get the exact measurements, a solution is to use some results from the research done on building façade images segmentation [18,19]. Thanks to such approaches, elements such as openings and balconies can be better recognized.

Moreover, some building elements are not recognized and included in the 3D building model, such as stairs and roofs. Stairs are usually represented on the floor plan by their footprint but no information is given about their shape. So it is possible to generate a bounding box around the stairs but not to generate automatically the corresponding 3D stairs. Another 3D reconstruction difficulty resides in the automatic generation of roofs which are essential to energy performances assessment. They are often represented by thin lines which cannot be easily recognized and isolated from other elements. Some works focus on the automatic generation of roofs from the footprint of the building or satellite images [20,21].

Finally, the 3D building models generated still contain some errors and shortcomings which cannot be automatically corrected, e.g. when the information is not available in the 2D floor plan. This is why another improvement perspective is to include some punctual human interventions during the recognition to check and complete the model when necessary in order to avoid the errors propagation. To do this, an analysis has to be done to describe errors and their inter-connections to optimize and minimize interventions.

5. Conclusion

This paper describes a methodology to automatically generate a 3D building model from a 2D scanned plan. 3D building models are intended to be used in the renovation domain for energy simulation for example. The process includes geometrical and topological corrections to provide a coherent 3D building model composed of walls, openings and spaces. A prototype of a software tool for 3D models generation from 2D plans has been developed and tested with a (90) plans database.

Two kinds of evaluation are outlined in this paper: a pixel-level evaluation and a building-level evaluation. Results are similar but the second evaluation seems to be more relevant because it reflects results according to the building and it is well understandable and interpretable for people working in the building construction domain. Walls are well recognized even if some building footprints are not validated. However, there is a large part of missing openings but only few false recognitions were observed, which is a very good point.

As part of our perspectives, in order to enrich and improve the 3D building model, we are planning to design and implement methods to enable assisted manual model-checking and correction. Another perspective is to extend the spectrum of the source data in order to enrich the 3D model. One very promising possibility is to rely on semantized building façades images and on aerial roof pictures.

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